

Software Installation Procedure

For the BrainWave® family of Products

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REVISION 8

August 14, 2008

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
1	May 3, 04	Pete Kargard	Initial release. This document was converted from the Brainwave SO manual.
2	May 17, 04	Pete Kargard	Feedback after FMI trial
3	June 14, 04	Kargard	Improvements after first in-house verification.
4	Sept 2005	Kargard	Updates for new computer
5	Sept 2005	Kargard	Update for new Boot sequence.
6	JAN 2006	Kargard	Changed all references to include release 12x and 14x.
7	May 2008	Kargard	Changed 'GEMS' to 'GE' in header. Removed strikeouts on pages 4-17, 5-20, 7-34, and 7-36 of Rev 6
8	August 14, 2008	K. Perella	Inserted Section 8: BrainWave on Signa MR750 3.0T with content for software installation procedure.

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1 Introduction

This manual describes software installation and system configuration for the following products.

Generic Product Name	Associated Catalog Number(s)
BrainWave on AW4.0	M10331BR: BrainWave AW4.0 license, operator manual, etc. • Always coupled with either full BrainWave Hardware or BrainWaveHW-Lite option (later in table). • RTIP option is a pre-requisite.
BrainWaveSO on AW4.0	M10331BZ: BrainWaveSO AW4.0 license etc. • RTIP option is a pre-requisite.
“Full” BrainWave Hardware Option Stimulus PC (legacy)	M1033BM: Stimulus PC plus magnet room audio, visual and patient response hardware. • No longer available to order – legacy installations only. • Compatible with both BrainWave (M10331BR) and BrainWaveRT (M1033BT). • See separate hardware installation (2301428-32) and troubleshooting documents (2301428-33).
BrainWaveHW-Lite Stimulus PC	M1033BL: Stimulus PC plus magnet room patient response hardware. • Compatible with both BrainWave (M10331BR) and BrainWaveRT (M1033BT). • See separate hardware installation (2402205) and troubleshooting (2402206) documents.

Important Notes:

- BrainWaveRT (M1033BT) and BrainWavePA (M1033BS) installations are not addressed in this document. These Excite scanner-based options are installed via the standard guided install process.
- An AW4.0 system may have either BrainWave or BrainWaveSO installed, not both.
- When loading a new version of AW4.0 BrainWave, or BrainWaveSO, be sure to run the uninstall procedure first (see “Uninstall BrainWave and BrainWaveSO Software” in the table of contents).
- If you are re-installing or upgrading ...
 - Excite scanner software
 - AW4.0 software or
 - BrainWave stimulus PC software,
 ... be sure to save customer developed paradigms first or this information will be **lost** (see “Saving Paradigm Information” in the Pre-Installation Checklists).

2 Important Configuration Information

2.1 BrainWave and BrainWaveSO on AW4.0

BrainWave application software CD 5110767
BrainWaveSO application software CD 2337089

Scanner Software	AW Software	RTIP Software
cnv4.283B_M4_0140.c + protocols patch 2328352 rev 1 + Patch_CNV4_M4_08 / RTIP 020529	AW 4.0.x	2271261-4
9.1_M3 or later	AW 4.0.x	2271261-8
vh3_M4 or later (3T Eclipse)	AW 4.0.x	2271261-8
10.0_M4 or later (MGD/Excite)	AW 4.0.x	2271261-8

2.2 “Full” BrainWave Hardware and BrainWaveHW-Lite

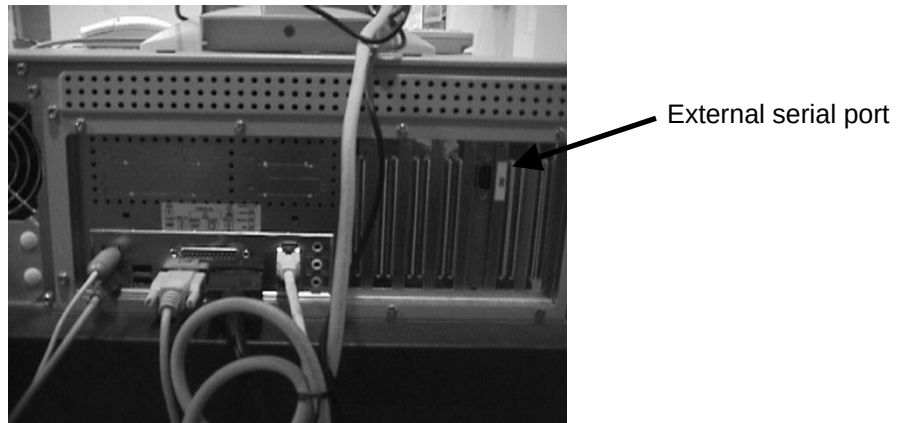
Both BrainWave hardware options (M1033BM and M1033BL) are compatible with BrainWave on AW4.0 (M10331BR) and BrainWaveRT (M1033BT). In all combinations, the stimulus PC application software is contained on CD part number **5110767**. It may also help to note that both the AW and Stimulus PC applications are delivered on this same CD (PC platform CD is separate).

Stimulus PC Hardware	Stimulus PC Platform Software
2303343-2	2303343-6
2303343-14	2303343-11
2394034-2	2394034-3
5158764	5158765

Important: The PC platform software revisions are tied to specific PC hardware parts as listed in the above table. The following *assemblies* consist of the appropriate pairings of Stimulus PC hardware and platform software. The stimulus PC *application* software runs correctly on both assemblies listed below.

PC + Platform SW Assembly	Possible Components
2303343-12 (FRU)	Either 2303343-2 (PC) + 2303343-6 (SW) -or- 2303343-14 (PC) + 2303343-11 (SW)
2309434-4 (FRU)	Either 2303343-2 (PC) + 2303343-6 (SW) -or- 2303343-14 (PC) + 2303343-11 (SW) -or- 2394034-2 (PC) + 2394034-3 (SW)
5158766	Contains 5158764 (PC) + 5158765 (SW)

To determine PC type, check to see if there is an external serial port on the back of the PC. If there is an external serial port, the part number of the PC is either 2303343-14 or 2394034-2. See Illustration 2-1. Also the new PCs will have the part number on a sticker in the front of the PC.



Part number 2303343-14 or 2394034-2 or 5158764

If the PC does not have the external serial port, it is the old PC or part number 2303343-2.



Part number 2303343-2

3 Manual Organization

From this point on, this manual really consists of four different sub-manuals for each of the following configurations.

- [Section 4] BrainWave on Excite **with** BrainWave Hardware
- [Section 5] BrainWave on Excite **without** BrainWave Hardware
- [Section 6] Lx scanner with **AW4.0** BrainWave
- [Section 7] Lx scanner with **AW4.0** BrainWaveSO
- [Section 8] BrainWave on Signa MR750 3.0T

Step-by-step instructions are provided for many subsequent procedures. Each step is presented in two ways, targeted separately for **Advanced Users** and **New Users**. Key pieces of information and operator actions are highlighted in **bold** for quick and easy execution.

4 BrainWave on Excite with BrainWave Hardware

This section covers systems with the following configuration:

Excite Scanner:	Software 11.0_M4, G3_M4, 12x, or 14x
BrainWaveRT scanner option:	Catalog M1033BT
BrainWavePA scanner option:	Catalog M1033BS - optional
BrainWave hardware option:	BrainWaveHW (M1033BL)

4.1 Pre-Installation Checklist

Record the appropriate information in the area at the bottom of the page.

4.1.1 License Key Check

Check that the following scanner license keys are installed or available depending on the options purchased on the Linux PC. These keys are installed using the standard Guided Install process.

- “brnwvrt” catalog: M1033BT – BrainWave Real-Time Acquisition
- “brnwvpa” catalog: M1033BS (optional) – BrainWave Post Acquisition
- “brnwvhw” catalog: M1033BL – BrainWave Hardware “Lite”

Locate the license key for Paradigm Development application on the Stimulus PC (NeuroActivator-Research).

- This stimulus PC specific key is part of catalog M1033BL and will often be found on a label attached to the BrainWave Stimulus PC.

4.1.2 Pre-Installation Information Record

Stimulus PC IP Address (e.g. 10.0.1.9) _____
see /etc/hosts entry for “gestimpc”

Stimulus PC Netmask: use this value -> 255.255.252.0

“NeuroActivator-Research” License Key (found on front of stimulus PC): _____

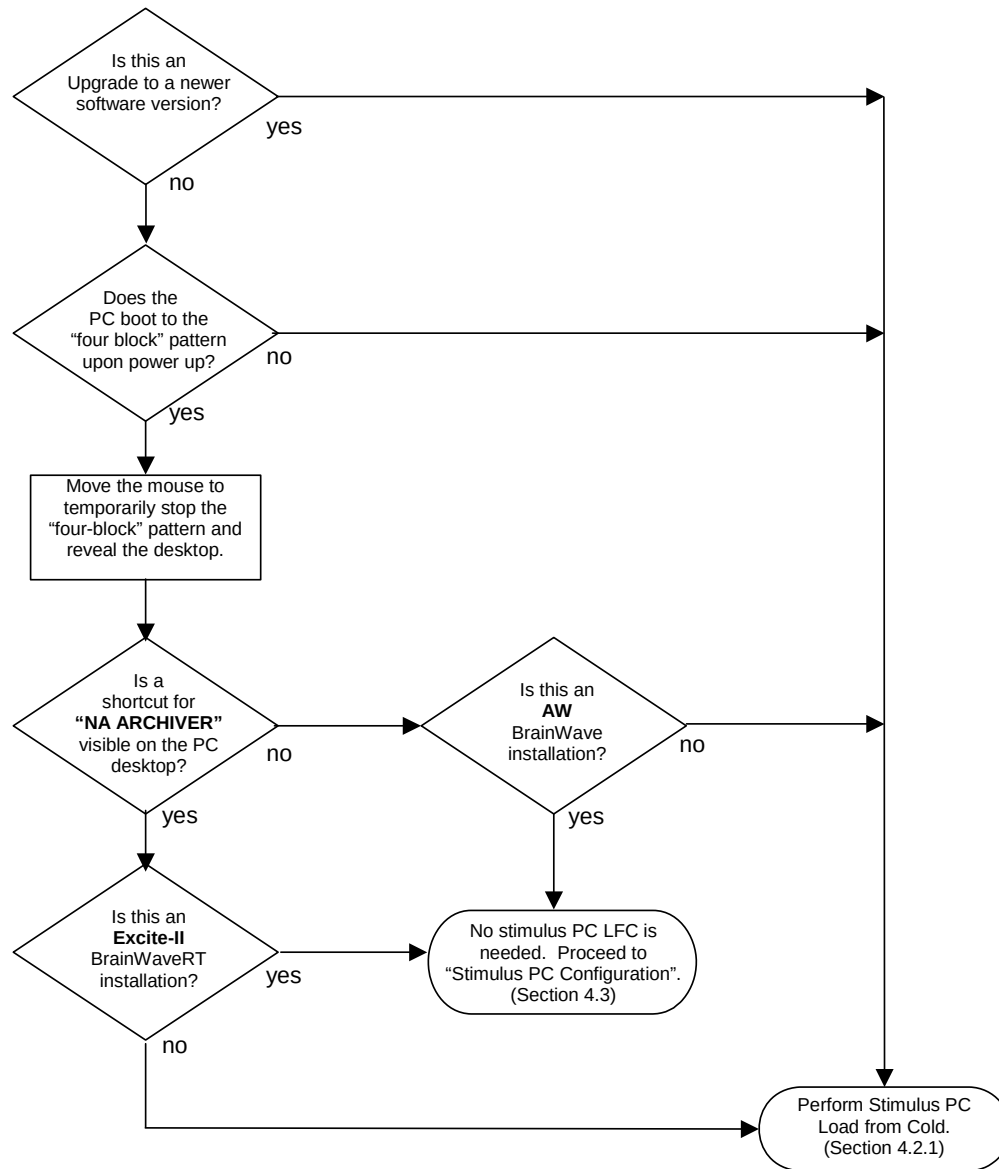
4.1.3 Saving Paradigm Information with BrainWaveRT on Excite

If you are upgrading or reloading either Excite scanner software or the Stimulus PC associated with the BrainWave Hardware option, be sure to save configuration data and any customer developed paradigms before reloading software. The customer generated paradigms are located in the research directory on the Stimulus PC.

Advanced Users	New Users
Service -> Utilities -> Paradigm Manager -> Start ...	1. Launch the Paradigm Manager from the Utilities Menu within the Service Desktop on the Excite Scanner console.
Hardware Paradigms	2. Select the "Hardware Paradigms" tab.
Open Window ... Right click to Exit NeuroActivator - Clinical	3. Press the " Open Window to Stim PC " button and right click over the NeuroActivator-Clinical icon that appears on the taskbar (lower right corner of screen) once the mouse is moved into the panel. Select Exit .
NA Archiver	4. Launch the NeuroActivator-Archiver. Use the save function to copy paradigm information (in a .zip file) from the Stimulus PC to the Excite scanner where it will be accessible to SaveInfo.
File -> Exit	5. Exit NA Archiver.
NeuroActivator - Clinical	6. Re-launch NeuroActivator-Clinical using the desktop shortcut. Verify that the four-block pattern returns once the mouse is outside the Stim PC window.
SaveInfo	7. Perform the standard SaveInfo function on the Excite scanner to insure all custom paradigm and configuration data has been saved.

4.2 Stimulus PC Load from Cold (15 minutes)

The Stimulus PC should normally arrive pre-loaded with both platform and application software compatible with Excite scanners. Perform the following checks to determine if reloading the Stimulus PC is necessary.



4.2.1 Stimulus PC Platform Installation

This steps loads a fresh image of the Stimulus PC operating system on the disk. At this time the platform consists of Windows NT Version 4 with Service Pack 6. Refer to Section 2.2 to determine needed software for Stimulus PC.

Advanced Users	New Users
Insert PC Platform CDROM	1. This CDROM may have one of several labels depending on your PC part number (see section 2). • (2303343-6) “NT Recovery Disk for BrainWave ...” • (2303343-11) or (5158765) “BrainWave Stimulus PC Disk Image (for BW Lite)” • (2394034-3) “BrainWave Stimulus PC Disk Image”
Start -> Shutdown (w/ auto restart)	2. Shutdown the PC with automatic restart selected. Note: Newer versions of Brainwave may require you to “Hit any key to start”. Hit the spacebar to continue
Autoload occurs	3. The system will auto-boot from the CD and load the software. Wait a few minutes until the following message appears, “ The software installation is complete. Please remove the CD and reboot. ”
Remove CD	4. Remove the CD from the CD-ROM drive.
CTRL-ALT-DEL	5. Press the CTRL-ALT-DEL keys simultaneously to reboot the PC . The PC will boot all the way to the standard start screen with no need for any additional operator intervention (i.e. no login required).

4.2.2 VNC Installation

VNC is the software application that permits remote display of the stimulus PC environment from either the AW (for BrainWave) or from the scanner console (for BrainWaveRT plus the BrainWave hardware option).

Advanced Users	New Users
Insert BrainWave CD	1. Insert the BrainWave application CDROM (5110767) into the Stimulus PC’s CD-ROM drive.
Windows Explorer	2. Start the Windows Explorer from the Start menu.
Select VNC	3. Use NT Explorer to go to the <CDROM drive>:\VNC directory
Setup.exe	4. Double click the Setup.exe (or “Setup”) icon.
Ok	5. Click OK in the warning message dialog box.

Advanced Users	New Users
Next	6. Click Next in the Welcome dialog box.
Yes	7. Click Yes in the SLA dialog box.
Next	8. Click Next. Leave the default destination folder.
Next	9. Click Next. Leave the default Program Folder entry.
Finish	10. Click Finish. The system may automatically reboot at this point.
<u>Install</u> WinVNC Service	11. Start -> Programs -> VNC -> Administrative Tools -> <u>Install</u> WinVNC Service
Ok	12. Click the "Ok" in the confirmation of service <u>installation</u> box. (This will disappear automatically after a few seconds.)
<u>Start</u> WinVNC Service	13. Start -> Programs -> VNC -> Administrative Tools -> <u>Start</u> WinVNC Service. (If you get an error, return to step 11 and <u>reinstall</u> .)
Ok	14. Click OK in the Warning dialog box.
Password = neuroactivator	15. Enter neuroactivator in the password field. (Caution: You <u>must</u> use this password.)
Ok	16. Click Ok.

4.2.3 Stimulus PC Application Software Install

These instructions are used to load Stimulus PC application software (NeuroActivator – Clinical, NeuroActivator – Research and NeuroArchiver).

Advanced Users	New Users
Insert BrainWave application CD.	1. Insert BrainWave 1.x application CD if not done already. The GE part number is 5110767.
TRIPLE CHECK !!!	WARNING: PLAY CLOSE ATTENTION TO THE NEXT STEP. THE BRAINWAVE APPLICATION CD SUPPORTS MULTIPLE CONFIGURATIONS. YOU MUST FOLLOW THE INSTRUCTIONS BELOW EXACTLY.
Use NT Explorer to find: BrainWaveRT_on_Scanner double click	2. Use NT Explorer to select folder <CDROM drive>:/ BrainWaveRT_on_Scanner Double click on this folder to view all the files contained there.

Advanced Users	New Users
Find Setup.exe double click	3. In the folder selected previously, you should see an icon labeled Setup or Setup.exe . Double click on this icon. This launches the NeuroActivator Installation Wizard .
Next	4. Quickly read the material in the first page of the Installation Wizard, then click on the Next button at the page bottom.
Next	5. The next page allows you to choose the folder into which NeuroActivator will be installed. You should accept the default location. Then click on the Next button.
Select all components Next Next	6. Insure all components in the Select Components menu are selected. Then click on the Next button and the second Next button to proceed with the installation. A status bar will fill from 0% to 100% as installation runs to completion.
Finish	7. In the final page click on the Finish button to exit the Installation Wizard .
Start -> Programs -> Neuroactivator -> NeuroActivator - Research	8. Activate the BrainWave paradigm editor application (NeuroActivator – Research).
Record HostID	9. Record the Host ID: _____
Find License Key	10. The application license key for NeuroActivator-Research should have been shipped with the BrainWave hardware option. On some Stimulus PCs, this key will appear on a label attached to the PC. If you are unable to find this license key, contact headquarters service engineering with the Host ID recorded above and the customer order number that was used to purchase the BrainWave hardware option. Password should be entered using lower case letters.
Enter Key	11. Enter the license key.
Exit	12. Exit NeuroActivator research version.
Remove CD ROM	13. It is safe to remove the installation CD at this point.

4.2.4 Stimulus PC Security Patch Installation

This step is required and should be the last step of the stimulus PC software installation process.

Advanced Users	New Users
Stop NeuroActivator – Clinical	1. Right click on the NeuroActivator Icon (black icon with an outline of a head located in lower right hand corner of monitor) and select “Exit”.
Insert Security Patches CD	2. Insert the BrainWave Stimulus PC Security Patches CDROM. GE Part Number: 5110968 .
Start -> Windows Explorer	3. Open the Windows Explorer and select the folder for the CDROM drive. Two files should be visible on this CD.
Double click NT_Patch_WB_...	4. Execute the Worm-Blaster virus patch executable (NT_Patch_WB_Q823980i.exe).
Remove CD and [OK]	5. Remove the Patch CDROM and then press the OK button which will cause the PC to reboot.
Repeat 1-5 for NT_Patch_RPC_...	6-10. Repeat steps 1-5 for the NT_Patch_RPC_K8824146.exe RPC virus patch.

4.3 Stimulus PC Configuration (10 minutes)

The same basic instructions are used to configure the Stimulus PC for both of the following configurations.

4.3.1 Stimulus PC Network Configuration

Advanced Users	New Users
Exit NeuroActivator	1. Move mouse to dismiss the flashing square. Right click on the NeuroActivator icon (image of the brain in the lower right hand corner of the task bar) and click exit.
Start->Settings->Control Panel -> Network	2. Launch the WinNT Network configuration utility. Go to Start -> Settings > Control Panel -> (double click “Network” icon)
Protocols	3. Select the Protocols tab.
Select TCP/IP	4. Select the TCP/IP protocol.
Properties	5. Select the Properties button.

Advanced Users	New Users
IP Address: <gestimpc IP addr> Subnet Mask: 255.255.252.0 Default Gateway: <leave blank>	6. From the tools desktop of the scanner workstation start a “ C Shell... ”. Using the “ more /etc/hosts ” command record the IP addresses for gestimpc _____. <site_hostname>-tgate _____. (used later) Enter the IP address assigned to “gestimpc” in the scanner hosts file and netmask (e.g. 255.255. 252 .0) reserved for the stimulus PC.
Ok	7. Click Ok in the TCP/IP Properties dialog box.
Ok	8. Click Ok in the Network dialog box.

4.3.2 NeuroActivator Configuration

Advanced Users

New Users

Advanced Users	New Users
Start -> Programs -> Windows Explorer	1. Use Windows Explorer to find and select folder ... C:\ProgramFiles\Sensor Systems, Inc.\NeuroActivator 1.x. w. Archiver 1.x IMPORTANT: Do not use the file below: C:\ProgramFiles\Sensor Systems, Inc.\NeuroActivator 1.x This will not allow communication between Host & StimPC during fMRI scans
Launch NeuroActivator_ Config.exe	2. In the NeuroActivator folder, you will see an icon labeled NeuroActivator_Config.exe . Double click on this icon to invoke the configuration program. The program asks for information about the Host Name, the IP address of the host, the Port (i.e., the network connection to the BrainWave console), and the local folder path on the PC where the stimulus program will store log files for data acquisition sessions.

Advanced Users	New Users
Hostname: <site_hostname>-tgate IP Address: <tgate-IP_address>	3. Enter the hostname and IP address of the Excite network. Enter the hostname and IP address that the stimulus PC can use to talk to the Excite scanner. This information can be located on the Excite scanner workstation by opening a “C Shell” through the tools desktop and typing in the command “more /etc/hosts” and noting the IP address and hostname associated with the “<site_hostname>-tgate” entry. For BrainWaveRT configurations, the Stimulus PC network cable is connected to the network hub located in the <i>Excite electronics cabinet</i> . Connect the network cable between the Stimulus PC and Hub in SC cabinet at this time.
Set Port to 3000	4. The port value should be set to 3000 . This value must match the port setting used later when performing BrainWave host configuration (either BrainWave on AW4.0 or BrainWaveRT on Excite).
Accept default and click Save	5. Accept the default location for the log files, C:\StimulusLogFiles . Next, click the Save option in the File menu.
Reboot? – Yes and then Exit Confirm that “four-block” pattern appears.	7. After clicking the Save option in the File menu, a dialog box will appear asking whether to reboot the PC when it exits. Click on the Yes button in the dialog box. Then quit the configuration program by selecting Exit under File . The PC will then reboot. NeuroActivator will automatically start and a flashing box will appear. This “screen-saver” is displayed to reduce the visual distractions the patient might see through the goggles while stimuli are not being delivered. Input from the mouse or keyboard will temporarily de-blank the screen.
	8. For the proper functioning of the BrainWave system, NeuroActivator (clinical version) must be running and the flashing box displayed.*

*If you need to gain control of the system, exit NeuroActivator by right-clicking the NeuroActivator icon (black background with outline of a head in profile) in the system tray visible in the lower right corner of the screen on the taskbar. In the popup menu, select **Exit** to stop the NeuroActivator application.

4.4 Postscript Printer Installation for BrainWavePA on Excite

Below is the procedure for installing the Codonics Horizon/PostScript printer for use with BrainWave PA.

In the following steps, you are to type the characters which appear in **boldface**.

1. Determine if Codonics Horizon/PostScript printer has already been installed:

```
% lpq -s  
horizon-cvp@t18 0 jobs (dest 7.a-cvp@horizon)
```

If the Codonics Horizon/PostScript does not appear in the output list then proceed to step 2.

2. Install the Codonics Horizon/PostScript printer

Before you begin you will need the hostname and IP address of the printer from the site network administrator (or Codonics Horizon Imager User's Manual for viewing printer network settings).

```
% su  
Password: <root password>  
[root@t18]# ~sdc/install/install.printer  
Installing hard copy device driver  
Is there a postscript printer available for your Workstation (y/n) : y  
Host name of the color printer : <printer hostname>  
Internet (IP) address of the color printer : <printer IP address>  
Remote queue name (for Horizon), otherwise use the default (lp) : 7.a-cvp  
Starting lpd: [ OK ]  
<...intermediate output deleted for clarity>  
Printer installation completed
```

Note: The remote queue name 7.a-cvp is specific to the Codonics Horizon Imager.

- 3 Verify you are able to print motion correction plots and Visualizer images from BrainWavePA.

If you are unable to print to the Codonics Horizon Imager from BrainWave PA, you may not have the PostScript feature enabled. Consult the Codonics Horizon Imager PostScript Technical Brief (http://www.codonics.com/pdf/901-068-002.a_horizon_ps.pdf – pertinent section copied below).

Checking Whether PostScript Is Enabled

1. Press the imager's **MENU** key.
2. Use the arrow keys to scroll to Test Prints and press the **ENTER** key.
3. Select an available media type and size.
4. Scroll to Imager Status and press the **ENTER** key. Press the **ENTER** key again to start one copy of the status print.
5. Press the **MENU** key to exit the menus.

After the test print is completed, examine it. Under the heading **PrinterFeatureInfo**, **Postscript** will be listed if it is available. The **Feature** status to the right will indicate **ENABLED** and there will be an indication as to whether the **PostScript** printing option is permanent or temporary, with an expiration date if temporary.

If **PostScript** printing is not enabled, contact your authorized Codonics Sales Representative.

5 BrainWave on Excite without BrainWave Hardware

This section covers systems with the following configuration:

- Excite Scanner: Software 11.0_M4 or G3_M4
- BrainWaveRT scanner option: Catalog M1033BT
- BrainWavePA scanner option: Catalog M1033BS - optional

5.1 License Key Check

Check that the following scanner license keys are installed or available depending on the options purchased. These keys are installed using the standard Guided Install process.

- “brnwvrt” catalog: M1033BT – BrainWave Real-Time Acquisition
- “brnwvpa” catalog: M1033BS (optional) – BrainWave Post Acquisition

5.2 Pre-Installation Checklist

If software is being reloaded or upgraded, the only pre-installation activities will be saving any software paradigms the user may have created and properly configuring the system to support postscript printing if an appropriate printer is available.

5.2.1 Saving Paradigm Information with BrainWaveRT on Excite

If you are upgrading or reloading the Excite scanner software be sure to save configuration data and any customer entered software paradigms before reloading software.

Advanced Users	New Users
SaveInfo	Perform the standard SaveInfo function on the Excite scanner to insure all custom paradigm and configuration data has been saved.

5.3 Postscript Printer Installation for BrainWavePA on Excite

Below is the procedure for installing the Codonics Horizon/PostScript printer for use with BrainWave PA.

In the following steps, you are to type the characters which appear in **boldface**.

1. Determine if Codonics Horizon/PostScript printer has already been installed

```
% lpq -s
horizon-cvp@t18 0 jobs (dest 7.a-cvp@horizon)
```

If the Codonics Horizon/PostScript does not appear in the output list, proceed to step 2.

2. Install the Codonics Horizon/PostScript printer

Before you begin, you will need the hostname and IP address of the printer from the site network administrator (or Codonics Horizon Imager User's Manual for viewing printer network settings).

```
% su
Password: <root password>
[root@t18]# ~sdc/install/install.printer
Installing hard copy device driver
Is there a postscript printer available for your Workstation (y/n) : y
Host name of the color printer : <printer hostname>
Internet (IP) address of the color printer : <printer IP address>
Remote queue name (for Horizon), otherwise use the default (lp) : 7.a-cvp
Starting lpd: [ OK ]
<...intermediate output deleted for clarity>
Printer installation completed
```

NOTE: The remote queue name 7.a-cvp is specific to the Codonics Horizon Imager.

3. Verify you are able to print motion correction plots and Visualizer images from BrainWavePA.

If you are unable to print to the Codonics Horizon Imager from BrainWave PA, you may not have the PostScript feature enabled. Consult the Codonics Horizon Imager PostScript Technical Brief (http://www.codonics.com/pdf/901-068-002.a_horizon_ps.pdf – pertinent section copied below).

Checking Whether PostScript Is Enabled

1. Press the imager's **MENU** key.
2. Use the arrow keys to scroll to Test Prints and press the **ENTER** key.
3. Select an available media type and size.
4. Scroll to Imager Status and press the **ENTER** key. Press the **ENTER** key again to start one copy of the status print.
5. Press the **MENU** key to exit the menus.

After the test print is completed, examine it. Under the heading **PrinterFeatureInfo**, **Postscript** will be listed if it is available. The **Feature** status to the right will indicate **ENABLED** and there will be an indication as to whether the **PostScript** printing option is permanent or temporary, with an expiration date if temporary.

If **PostScript** printing is not enabled, contact your authorized Codonics Sales Representative.

6 Lx scanner with AW4.0 BrainWave

This section covers systems with the following configuration:

Lx Scanner: Software cnv4_M4, 9.1_M4, or vh3_M4 (or later)
 BrainWave AW4.0 option: Catalog M10331BR
 BrainWave hardware option: Catalog M1033BL (or legacy number M1033BM)

6.1 Pre-Installation Checklist

6.1.1 Check AW4.0 Installation

Verify that you have an Ultra-60 based AW with software revision AW4.0. BrainWave is not compatible with AW versions 4.1 and beyond. Use the AW “Display Configuration” function to verify the software revision and all installed options.

6.1.2 RTIP Check

Verify that the RTIP option is properly installed. The Ultra 60 must have a second SCSI controller card installed which is connected to a separate disk or disk tower. Perform the Unix command “df -k” in a terminal window and verify there is a separate partition in the list with “/export/RTIP” as the associated directory.

A common error occurs if performing an AW4.0 load from cold (LFC) with the RTIP SCSI disk turned on. An AW4.0 LFC **MUST** be performed with power removed from the RTIP SCSI disk. Power is turned on just before the RTIP software installation.

6.1.3 Record Site Specific Information

AW4.0 Workstation IP Address: ____ . ____ . ____ . ____ Hostname: _____

BrainWave AW Application License Key: _____

Stimulus PC IP Address: ____ . ____ . ____ . ____ Netmask: ____ . ____ . ____ . ____

(Note: The Stimulus PC IP address and netmask must be assigned by the customer’s network administrator, and must be on the same subnet as the Lx scanner and AW4.0 workstation that will be used by BrainWave.)

Note

The NeuroActivator Research License Key is an option, may not be present for all sites.

NeuroActivator-Research License Key: _____

(Note: The PC-specific license key is often found on a label attached to the stimulus PC or with documentation associated with the BrainWave HW option. If this information is not available, record the PC Host ID that is displayed later in the procedure when first attempting to run the “NeuroActivator – Research” application on the stimulus PC.)

(optional) Color Postscript® printer IP address and AW4.0 root password.

Postscript Printer IP Address: ____ . ____ . ____ . ____

AW4.0 Root Password: _____

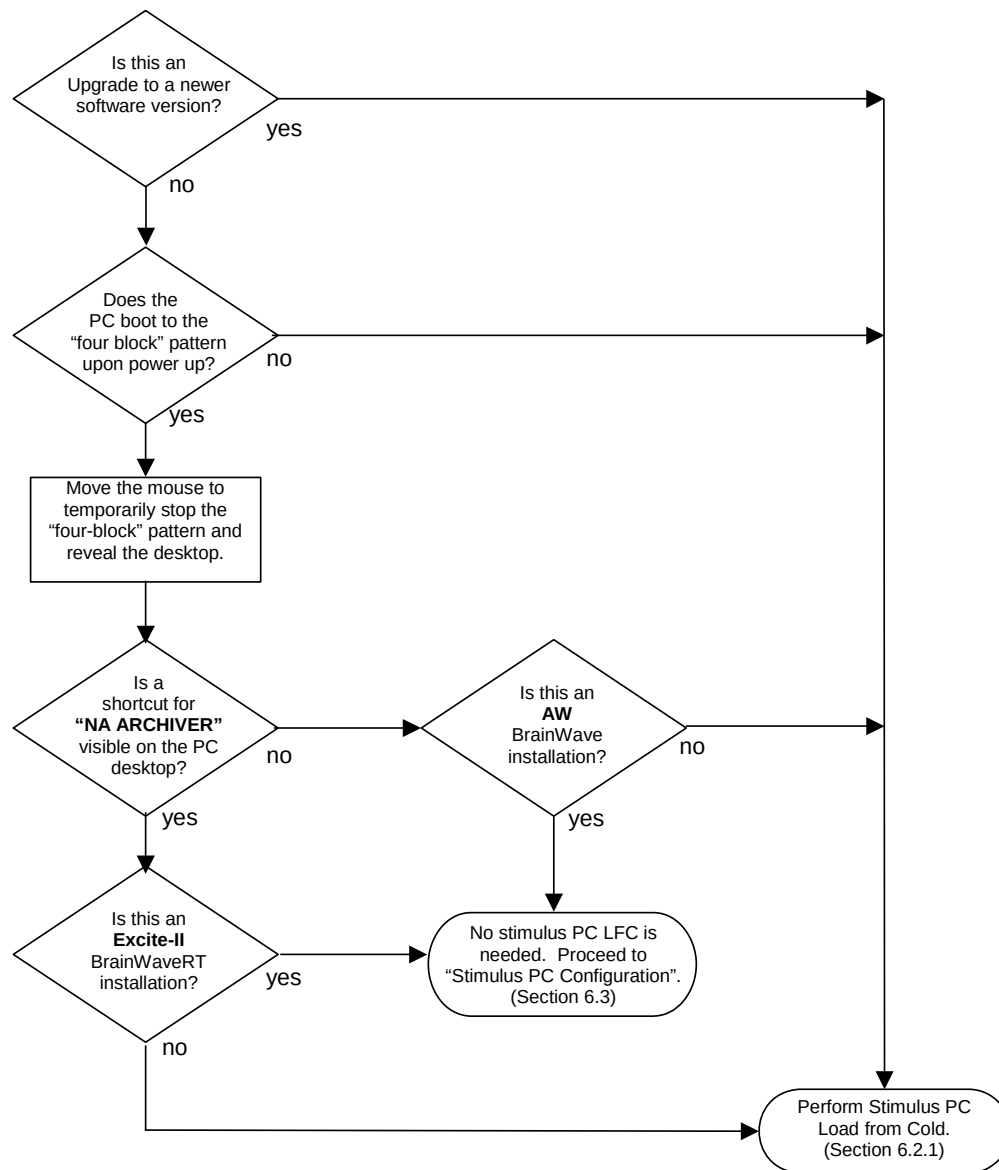
6.1.4 Saving Paradigm Information with BrainWave on AW4.0

The following procedure should be used to back up any customer developed BrainWave paradigms prior to reloading BrainWave software on the AW4.0 workstation or reloading the stimulus PC. Any Customer developed paradigms are located in the research directory on the stimulus PC. This step is not needed if the customer has not generated any custom paradigms.

Advanced Users	New Users
Stop NeuroActivator – Clinical	1. Right click on the NeuroActivator Icon (black icon with an outline of a head) and select “Exit”.
Start -> Run -> cmd	2. Open a command window on the stimulus PC.
ftp	3. Use (binary) ftp to retrieve file ~sdc/sensor/nn_db/ParadigmInfo.txt onto the Windows desktop.
Insert blank CDROM	4. Insert a blank CDROM into the stimulus PC CDRW drive.
Save key info to CDROM	5. Using the CD Writer software provided on the Stimulus PC, save the ParadigmInfo.txt file, plus the complete contents of the NeuroActivator “Research” directory. Remember that the .stm files are ASCII files containing only some of the information associated with each paradigm. Be sure to save all associated images, and sound files as well (usually placed in subdirectories of the “Research” directory by convention. Be sure to consult with the customer to insure all important files have been saved.

6.2 Stimulus PC Load from Cold (15 minutes)

The Stimulus PC should normally arrive pre-loaded with both platform and application software. Perform the following checks to determine if reloading the Stimulus PC is unnecessary.



6.2.1 Stimulus PC Platform Installation

This step loads a fresh image of the Stimulus PC operating system on the disk. At this time the platform consists of Windows NT Version 4 with Service Pack 6. See Section 2-2 for different Stimulus PC configurations.

Advanced Users	New Users
Insert PC Platform CDROM	6. This CDROM may have one of several labels depending on your PC part number (see section 2). • (2303343-6) “NT Recovery Disk for BrainWave ...” • (2303343-11) “BrainWave Stimulus PC Disk Image” • (2394034-3) “BrainWave Stimulus PC Disk Image”
Start -> Shutdown (w/ auto restart)	7. Shutdown the PC with automatic restart selected.
Autoload occurs	8. The system will auto-boot from the CD and load the software. Wait a few minutes until the following message appears, “ The software installation is complete. Please remove the CD and reboot. ” Platform load will take about 5 minutes.
Remove CD	9. Remove the CD from the CD-ROM drive.
CTRL-ALT-DEL	10. Press the CTRL-ALT-DEL keys simultaneously to reboot the PC . The PC will boot all the way to the standard start screen with no need for any additional operator intervention (i.e. no login required).

6.2.2 VNC Installation

VNC is the software application that permits remote display of the stimulus PC environment from either the AW (for BrainWave) or from the scanner console (for BrainWaveRT plus the BrainWave hardware option).

Advanced Users	New Users
Insert BrainWave CD	1. Insert the BrainWave application CDROM 5110767 into the Stimulus workstation's CD-ROM drive.
Windows Explorer	2. Start the Explorer from the Start menu.
Select VNC	3. Use Windows Explorer to navigate to the <CDROM drive>:\VNC directory
Setup.exe	4. Double click the Setup.exe (or “Setup”) icon.
Ok	5. Click OK in the warning message dialog box.
Next	6. Click Next in the Welcome dialog box.

Advanced Users	New Users
Yes	7. Click Yes in the SLA dialog box.
Next	8. Click Next. Leave the default destination folder.
Next	9. Click Next. Leave the default Program Folder entry.
Finish	10. Click Finish. NOTE: The system may automatically reboot at this point.
<u>Install</u> WinVNC Service	11. Start -> Programs -> VNC -> Administrative Tools -> <u>Install</u> WinVNC Service
Ok	12. Click the "Ok" in the confirmation of service <u>installation</u> box. (This will disappear automatically after a few seconds.)
<u>Start</u> WinVNC Service	13. Start -> Programs -> VNC -> Administrative Tools -> <u>Start</u> WinVNC Service. (If you get an error, return to step 11 and <u>reinstall</u> .)
Ok	14. Click OK in the Warning dialog box.
Password = neuroactivator	15. Enter neuroactivator in the password field. (Caution: You <u>must</u> use this password.)
Ok	16. Click Ok.

6.2.3 Stimulus PC Application Software Install

The same basic instructions are used to load Stimulus PC application software for both of the following configurations.

Advanced Users	New Users
Insert BrainWave CD.	1. If not already done, insert BrainWave application CD (GE part number 5110767.).
TRIPLE CHECK !!!	WARNING: PLAY CLOSE ATTENTION TO THE NEXT STEP. THE BRAINWAVE APPLICATION CD SUPPORTS MULTIPLE CONFIGURATIONS. YOU MUST FOLLOW THE INSTRUCTIONS BELOW EXACTLY.
Find folder... BrainWave_RTIP_on_AW double click	2. Use Windows Explorer to select folder ... <CDROM drive>:/BrainWave_RTIP_on_AW Prepare to install the stimulus PC application compatible with BrainWave on AW4.0.
Find Setup.exe double click	3. In the folder selected previously, you should see an icon labeled Setup or Setup.exe . Double click on this icon. This launches the NeuroActivator Installation Wizard .

Advanced Users	New Users
Next	4. Quickly read the material in the first page of the Installation Wizard, then click on the Next button at the page bottom.
Next	5. The next page allows you to choose the folder into which NeuroActivator will be installed. You should accept the default location. Then click on the Next button.
Select all components Next Next	6. Insure all components in the Select Components menu are selected. Then click on the Next button and the second Next button to proceed with the installation. A status bar will fill from 0% to 100% as installation runs to completion.
Finish	7. In the final page click on the Finish button to exit the Installation Wizard .
Start -> Programs -> Neuroactivator -> NeuroActivator - Research	8. Activate the BrainWave paradigm editor application (NeuroActivator – Research).
Record HostID	9. Record the Host ID: 000347950F42 (example)
Find License Key	10. The application license key for NeuroActivator-Research should have been shipped with the BrainWave hardware option. On some Stimulus PCs, this key will appear on a label attached to the PC. If you are unable to find this license key, contact headquarters service engineering with the PC Host ID recorded in the previous step and the customer order number that was used to purchase the BrainWave hardware option.
Enter Key	11. Enter the license key.
Exit	12. Exit NeuroActivator research version.
Remove CD ROM	13. It is now safe to remove the installation CD.

6.2.4 Stimulus PC Security Patch Installation

This step is required and should be the last step of the stimulus PC software installation process.

Advanced Users	New Users
Stop NeuroActivator – Clinical	1. Right click on the NeuroActivator Icon (black icon with an outline of a head) and select “Exit”.
Insert Security Patches CD	2. Insert the BrainWave Stimulus PC Security Patches CDROM. GE Part Number: 5110968.

Advanced Users	New Users
Start -> Windows Explorer	3. Open Windows Explorer and select the folder for the CDROM drive. Two files should be visible on this CD.
Double click NT_Patch_WB_...	4. Execute the Worm-Blaster virus patch executable (NT_Patch_WB_Q823980i.exe).
Remove CD and [OK]	5. Remove the Patch CDROM and then press the OK button which will cause the PC to reboot.
Repeat 1-5 for NT_Patch_RPC_...	6-10. Repeat steps 1-5 for the NT_Patch_RPC_K8824146.exe RPC virus patch.

6.3 Stimulus PC Network Configuration

Advanced Users	New Users
Exit NeuroActivator	1. Move mouse to dismiss the flashing square. Right click on the NeuroActivator icon (image of the brain in the lower right hand corner of the task bar) and click exit.
Start->Settings-> Control Panel -> Network	2. Launch the WinNT Network configuration utility. Go to Start -> Settings > Control Panel -> (double click "Network" icon)
Protocols	3. Select the Protocols tab.
Select TCP/IP	4. Select the TCP/IP protocol.
Properties	5. Select the Properties button.
IP Address: Subnet Mask: Default Gateway:	6. Enter the <i>site assigned IP address, subnet mask and (optional) gateway</i> recorded on the pre-installation checklist earlier in this document. This is the IP information the AW will use to talk to the stimulus PC through the network hub located in the scanner's duplex cabinet.
Ok	7. Click Ok in the TCP/IP Properties dialog box.
Ok	8. Click Ok in the Network dialog box.

6.3.1 NeuroActivator Configuration

Advanced Users	New Users
Start -> Programs -> Windows Explorer	1. Use Windows Explorer to find and select folder ... C:\ProgramFiles\Sensor Systems, Inc.\NeuroActivator 1.x.

Advanced Users	New Users
Launch NeuroActivator_ Config.exe	2. In the NeuroActivator folder, you will see an icon labeled NeuroActivator_Config.exe . Double click on this icon to invoke the configuration program. The program asks for information about the Host Name, the IP address of the host, the Port (i.e., the network connection to the BrainWave console), and the local folder path on the PC where the stimulus program will store log files for data acquisition sessions.
Enter the AW... Hostname: IP address:	3. Enter the hostname and IP address that the stimulus PC can use to talk to the AW workstation. This information can be located on the AW workstation via Tools -> Command Window and typing in the command “more /etc/hosts” and noting the IP address and hostname associated with the “loghost”. Note: For BrainWave configurations, the Stimulus PC network cable is connected to the network hub located in the scanner's <i>duplex cabinet</i> .
Set Port to 3000	4. The port value should be set to 3000 . This value must match the port setting used later when performing BrainWave host configuration (either BrainWave on AW4.0 or BrainWaveRT on Excite).
Accept default and click Save	5. Unless there are overriding reasons, you should accept the default location for the log files, C:\StimulusLogFiles . Next, click the Save option in the File menu.
Reboot? – Yes and then Exit Confirm that “four-block” pattern appears.	6. After clicking the Save option in the File menu, a dialog box will appear asking whether to reboot the PC when it exits. Click on the Yes button in the dialog box. Then quit the configuration program by selecting Exit under File . The PC will then reboot. NeuroActivator will automatically start and a flashing box will appear. This “screen-saver” is displayed to reduce the visual distractions the patient might see through the goggles while stimuli are not being delivered. Input from the mouse or keyboard will temporarily de-blank the screen.
	7. For the proper functioning of the BrainWave system, NeuroActivator (clinical version) must be running and the flashing box displayed.*

*If you need to gain control of the system, exit NeuroActivator by right-clicking the NeuroActivator icon (black background with outline of a head in profile) in the system tray visible in the lower right corner of the screen on the taskbar. In the popup menu, select **Exit** to stop the NeuroActivator application.

6.4 BrainWave Installation and Configuration

Note that this software is only compatible with AW4.0 (since the pre-requisite RTIP option is only compatible with AW4.0 as well).

6.4.1 AW4.0 Software Installation

Advanced Users	New Users
Log in as sd Tools-> Command Window	1. Log in as sd and open a command window. (Browser Tab: Tools-> Command Window) The AW system should be idle with no remote users accessing the system or trying to send data. Warning: Do not perform easyinstall as root. The AW system will be adversely affected resulting in the need to perform a full AW Software Load From Cold!
Check for and uninstall previous versions.	2. Check for a previous BrainWave or BrainWaveSO installation. Type the following in the command window: <pre>% cd % cd sensor (if directory does not exist then skip to step 3) ----- % su root # ./uninstall.brainwave (The leading "dot slash" is required.) # exit</pre> Be sure to exit root mode before continuing!
Insert BrainWave CDROM	3. Load the BrainWave CDROM (GE part 5110767). Note that this is the same CD used to load the Stimulus PC software and is delivered as part of the BrainWave Hardware option (either M1033BM or M1033BL).
Type easyinstall	Warning: Do not perform easyinstall as root. The AW system database will be adversely affected resulting in the need to perform a full AW Software Load From Cold! 4. Run easyinstall by typing "easyinstall" at the command prompt. The easyinstall installation scripts will guide you through the installation of the software.
Driver: CD-Rom	5. Select Driver: CD-Rom.
Contents	6. Click the Contents button.
Install	7. Click the install button.

Advanced Users	New Users
Enter license key	8. Enter the AW license key are recorded in the pre-installation checklist at the beginning of this document and hit return. The rest of the installation will proceed automatically. Examine the message area for the string “Unlimited License Installed” as evidence of a successful install. Possible errors are: insufficient disk space (free up disk space in the \$SdC_IMAGE_POOL directory and try installing again), invalid license key (try installing again and entering the key again or request a new key after checking the system information required for key generation is correct).
Eject CD-ROM	9. Click the Eject CD-ROM button.
Quit and Ok	10. Click the Quit button and then Ok to exit EasyInstall.
Exit	11. Type exit in the C-Shell window to terminate the C-Shell.

6.4.2 BrainWave Configuration for Solaris (AW)

Advanced Users	New Users
Security -> Restart Software	1. Shut down the AW browser and restart it (Browser Tab: Security -> Restart Software). A popup box will ask “Do you really want to restart the Software?”. Click yes.
More Software -> BrainWave	2. Launch the newly installed AW application. The AW browser will be automatically set aside, and the BrainWave graphical user interface (GUI) will appear.
Manage System	3. Press the “Manage System” button on the BrainWave main menu.
Modify Configuration Defaults and Set Modes Enter password Update RTIP path with first three letters of scanner suite name.	4. Press the “Modify Configuration Defaults and Set Modes” button. You will be prompted to enter a password. As this is the first time you are entering the password you are simultaneously setting the password. It is suggested that you establish the standard system root password already in use on the AW workstation. Under the Media Storage tab, check that the “Advantage Windows RTIP Directory” is set appropriately so BrainWave can find the RTIP disk. <code>/export/RTIP/idbm/database/MROR/GENESIS/<first_three_char_suite_name></code> All other default values should not be changed.

Advanced Users	New Users
Enter IP address Verify Port = 3000	5. Select the “Stimulus System” tab. Enter the IP address and port number of the Stimulus PC as established during pre-installation.
DICOM Enter AW hostname and IP Verify Port=4006	6. Select the DICOM tab. Enter hostname and IP address for the AW4.0 workstation. This information is used by BrainWave when installing results in the AW database. Verify port is set to 4006.
Set Modes... Main Menu	7. Exit the configuration panel by selecting “ Set Modes and Save Config Settings ”. Press “ Return to the Main Menu ” to exit the Manage panel.
Quit BrainWave Restart BrainWave	8. Exit BrainWave by pressing “ Quit BrainWave ” button, and then restart it through the AW Browser “More Software” function. Verify that BrainWave now starts with no BrainWave warnings posted.
NeuroActivator connected - confirm	9. The BrainWave Messages panel in BrainWave will now display a message NeuroActivator connected. In BrainWave's Manage System interface, click on the button labeled Check Status of NeuroActivator. The BrainWave Messages panel in BrainWave will then display a message NeuroActivator is in idle mode. If you do not see these messages in the BrainWave Messages panel, something is wrong with the configuration, NeuroActivator is not currently running, or there is a problem with the network.

6.5 Postscript® Printer Setup for AW4.0

This step is run on the AW Solaris Workstation. Currently only the Codonics color printer is supported. Text after the % should be typed in the command window. Text in ()'s are comments that should not be typed in. This same procedure applies to BrainWave and BrainWaveSO.

Advanced Users	New Users
sdc Tools-> Command Window	1. Log in as sdc and open a command window. (Browser Tab: Tools-> Command Window)
% su -root	2. % su -root (you will need to type in the root password for the AW workstation)
% ~sdc/install/install.printers	3. If you have a printer provided by GE then type % ~sdc/install/install.printers For other printers skip to step 12

Advanced Users	New Users
Return	4. If you have a Configuration Floppy you should insert it now. Regardless, press [Return] to continue ...
y	5. Do you want to install new printer(s) [y,n,?,q] y
GENNavprntr	6. Define the local name of the printer [?,q] GENNavprntr (Note: You must use this printer name)
Enter selection [?,??,q]: 1	7. Installing Printer Printer model 1 c Codonics (1660M 1660MD) 2 l Lexmark (Optra 1650N 1855N SC1275N C710n) 3 s Seiko (1720D)
Enter selection [?,??,q]: 1	8. Printer Type: 1 c Color printer 2 g Grayscale printer
Enter IP address	9. Printer internet (IP) address [?,q] ex. 192.168.1.101
Enter hostname	10. Printer hostname [?,q] ex. hp4050n Checking if hp4050n is alive ... Changing spool directory ... Print services stopped. Changing spool directory ... Print services started. Removing printer (if necessary) ... Declaring printer ... Setting filming configuration file ...
n	11. Do you want to install new printer(s) [y,n,?,q] n
Other printers	12. If you have a different printer, follow these steps.
sdcc Tools-> Command Window	13. Log in as sdc and open a command window. (Browser Tab: Tools-> Command Window)
% su -root	14. % su -root (you will need to type in the root password for the AW workstation)
Generic method	15. % lpset -a bsdaddr=<printer_name>,scaled,Solaris GENNavprntr
Example	16. lpset -a bsdaddr=PS_RAD1,scaled,Solaris GENNavprntr

7 Lx scanner with AW4.0 BrainWaveSO

This section covers systems with the following configuration:

Lx Scanner: Software cnv4_M4, 9.1_M4, or vh3_M4 (or later)

BrainWaveSO AW4.0 option: Catalog M10331BZ

7.1 Pre-Installation Checklist

7.1.1 Check AW4.0 Installation

Verify that you have an Ultra-60 based AW with software revision AW4.0. BrainWaveSO is not compatible with AW versions 4.1 and beyond. Use the AW “Display Configuration” function to verify the software revision.

7.1.2 RTIP Check

Verify that the RTIP option is properly installed. The Ultra 60 must have a second SCSI controller card installed which is connected to a separate disk or disk tower. Perform the Unix command “df -k” in a terminal window and verify there is a separate partition in the list with “/export/RTIP” as the associated directory.

A common error occurs if performing an AW4.0 load from cold (LFC) with the RTIP SCSI disk turned on. An AW4.0 LFC **MUST** be performed with power removed from the RTIP SCSI disk. Power is turned on just before the RTIP software installation.

7.1.3 Record Site Specific Information

AW4.0 Workstation IP Address: ____ . ____ . ____ . ____ Hostname: _____

BrainWaveSO AW Application License Key: _____

- (optional) Color Postscript® printer IP address and AW4.0 root password.

Postscript Printer IP Address: ____ . ____ . ____ . ____

AW4.0 Root Password: _____

7.1.4 Saving Paradigm Information with BrainWaveSO on AW4.0

If you are upgrading or reloading BrainWaveSO software on the AW4.0 workstation, be sure to save configuration and paradigm information prior to the operation.

The BrainWaveSO “Manage System” page includes a function to save BrainWaveSO storage directories, communications port, and DICOM settings as well as the list of Paradigms defined on the system and their color associations.

Advanced Users	New Users
More Software-> BrainWaveSO	1. Start BrainWaveSO if it is not already running. (Browser Tab: More Software-> BrainWaveSO.) The BrainWaveSO GUI will pop up.

Advanced Users	New Users
Manage System Save / Restore ... Enter password	2. Select Manage System . Select Save / Restore BrainWaveSO System Information . You will be prompted for the service password (see II-B step 3), enter it.
Save ... Ok	3. To save the system information to floppy disk select the Save Information to Floppy Disk button. You will be presented with a confirm box and prompted to insert a formatted* floppy disk. Do so and hit ok. Once disk activity stops eject the disk, label it, and store it in a safe location. The files saved are ~sdc/sensor/neuronav/config/NeuroNav.config and ~sdc/sensor/nn_db/ParadigmInfoSO.txt This saves the communications port, DICOM settings, and list of Paradigms and their color associations. *To format a floppy disk use the following command in a command window (unix prompt) : "fdformat -dfv"
Restore ... Ok Restart BrainWaveSO	4. To restore the system information from floppy disk select the Restore Information from Floppy Disk button. You will be presented with a confirmation box and prompted to insert a floppy disk containing previously saved system information. The files restored are ~sdc/sensor/neuronav/config/NeuroNav.config and ~sdc/sensor/nn_db/ParadigmInfoSO.txt. Backup copies of the files currently on disk (if any) will be made with the .bak extension. To ensure the settings have taken place exit and restart BrainWaveSO.

7.2 BrainWaveSO Installation and Configuration

Note that this software is only compatible with AW4.0 (since the pre-requisite RTIP option is only compatible with AW4.0 as well).

7.3 AW4.0 Software Installation

Advanced Users	New Users
Log in as sdC Tools-> Command Window	1. Log in as sdC and open a command window. (Browser Tab: Tools-> Command Window) The AW system should be idle with no remote users accessing the system or trying to send data. Warning: Do not perform easyinstall as root. The AW system will be adversely affected resulting in the need to perform a full AW Software Load From Cold!
Check for and uninstall previous versions.	2. Check for a previous BrainWave or BrainWaveSO installation. Type the following in the command window: <pre>% cd % cd sensor (if directory does not exist then skip to step 3) % su root # ./uninstall.brainwave (The leading "dot slash" is required.) # exit</pre> Be sure to exit root mode before continuing!
Insert BrainWaveSO CDROM	3. Load the BrainWaveSO application CD (GE part 2337089).
Type easyinstall	Warning: Do not perform easyinstall as root. The AW system database will be adversely affected resulting in the need to perform a full AW Software Load From Cold! 4. Run easyinstall by typing "easyinstall" at the command prompt. The easyinstall installation scripts will guide you through the installation of the software.
Driver: CD-Rom	5. Select Driver: CD-Rom.
Contents	6. Click the Contents button.
Install	7. Click the install button.
Enter license key	8. Enter the AW license key recorded in the pre-installation checklist at the beginning of this document and hit return. The rest of the installation will proceed automatically. Examine the message area for the string "Unlimited License Installed" as evidence of a successful install. Possible errors are: insufficient disk space (free up disk space in the \$SdC_IMAGE_POOL directory and try installing again), invalid license key (try installing again and entering the key again or request a new key after checking the system information required for key generation is correct).

Advanced Users	New Users
Eject CD-ROM	9. Click the Eject CD-ROM button.
Quit and Ok	10. Click the Quit button and then Ok to exit EasyInstall.
Exit	11. Type exit in the C-Shell window to terminate the C-Shell.

7.3.1 BrainWaveSO Configuration for Solaris (AW)

Advanced Users	New Users
Security -> Restart Software	1. Shut down the AW browser and restart it (Browser Tab: Security -> Restart Software). A popup box will ask "Do you really want to restart the Software?". Click yes.
More Software -> BrainWaveSO	2. Launch the newly installed AW application. The AW browser will be automatically set aside, and the BrainWaveSO graphical user interface (GUI) will appear.
Manage System	3. Press the "Manage System" button on the BrainWaveSO main menu.
Modify Configuration Defaults and Set Modes Enter password Update RTIP path with first three letters of scanner suite name.	4. Press the "Modify Configuration Defaults and Set Modes" button. You will be prompted to enter a password. As this is the first time you are entering the password you are simultaneously setting the password . It is suggested that you use the standard system root password already in use on the AW workstation. Under the Media Storage tab, check that the "Advantage Windows RTIP Directory" is set appropriately so BrainWave can find the RTIP disk. /export/RTIP/ldbm/database/MROR/GENESIS/<first_three_char_suite_name> All other default values should not be changed.
DICOM Enter AW hostname and IP Verify Port=4006	5. Select the DICOM tab. Enter hostname and IP address for the AW4.0 workstation. This information is used by BrainWaveSO when installing results in the AW database. Verify port is set to 4006.
Set Modes... Main Menu	6. Exit the configuration panel by selecting " Set Modes and Save Config Settings ". Press " Return to the Main Menu " to exit the Manage panel.

Advanced Users	New Users
Quit BrainWaveSO Restart BrainWaveSO	7. Exit BrainWaveSO by pressing “ Quit BrainWaveSO ” button, and then restart it through the AW Browser “More Software” function. Verify that BrainWaveSO now starts with no BrainWave warning messages posted.

7.4 Postscript® Printer Setup for AW4.0

This step is run on the AW Solaris Workstation. Currently only the Codonics color printer is supported. Text after the % should be typed in the command window. Text in ()'s are comments that should not be typed in. This same procedure applies to BrainWave and BrainWaveSO.

Advanced Users	New Users
sdc Tools-> Command Window	1. Log in as sdc and open a command window. (Browser Tab: Tools-> Command Window)
% su –root	2. % su –root (you will need to type in the root password for the AW workstation)
% ~sdc/install/install.printers	3. If you have a printer provided by GE then type % ~sdc/install/install.printers For other printers skip to step 12
Return	4. If you have a Configuration Floppy you should insert it now. Regardless, press [Return] to continue ...
y	5. Do you want to install new printer(s) [y,n,?,q] y
GENNavprntr	6. Define the local name of the printer [?,q] GENNavprntr (Note: You must use this printer name)
Enter selection [?,??,q]: 1	7. Installing Printer Printer model 1 c Codonics (1660M 1660MD) 2 l Lexmark (Optra 1650N 1855N SC1275N C710n) 3 s Seiko (1720D)
Enter selection [?,??,q]: 1	8 Printer Type: 1 c Color printer 2 g Grayscale printer
Enter IP address	9. Printer internet (IP) address [?,q] ex. 192.168.1.101

Advanced Users	New Users
Enter hostname	10. Printer hostname [?,q] ex. hp4050n Checking if hp4050n is alive ... Changing spool directory ... Print services stopped. Changing spool directory ... Print services started. Removing printer (if necessary) ... Declaring printer ... Setting filming configuration file ...
n	11. Do you want to install new printer(s) [y,n,?,q] n
Other printers	12. If you have a different printer, follow these steps.
sd c Tools-> Command Window	13. Log in as sdc and open a command window. (Browser Tab: Tools-> Command Window)
% su -root	14. % su -root (you will need to type in the root password for the AW workstation)
Generic method	15. % lpset -a bsdaddr=<printer_name>,scaled,Solaris GENNavprntr
Example	16. lpset -a bsdaddr=PS_RAD1,scaled,Solaris GENNavprntr

8 BrainWave on Signa MR750 3.0T

The drive image for BrainWave software is created by Granite Microsystems Inc. The software arrives pre-loaded and ready to use with the BrainWave hardware.

8.1 Software Setup

Advanced Users	New Users
Boot Computer	1. Once BrainWave hardware is completely assembled, turn on the Lumina box in front of the computer. This switch is located on the backside of the box. 2. Turn on computer. 3. Computer will boot. Software will automatically login and start. 4. Once system is displaying the "Idle" screen, proceed to Section 8.1.2: Checking Network PC Configuration.

8.1.1 Stop and Start PXE

Advanced Users	New Users
Access desktop momentarily.	To temporarily go to the Windows Desktop, press [Esc] . The PXE will automatically return to idle state after a time out.
Exit PXE	Press [Shift] + [Esc] to stop PXE and go to Windows Desktop.
Launch PXE	Double-click on Launch BrainWave PXE desktop icon to start PXE. The "Idle" screen will then display on the monitor.

8.1.2 Checking Network PC Configuration

Advanced Users	New Users
Exit PXE	On STIM PC, press [Shift] + [Esc] to stop PXE and go to Windows Desktop
Start > Control Panel > Network Connections	1. To view network properties, click Start on the Windows Desktop and select Control Panel. 2. Double-click on Network Connections. 3. Right-click on Local Area Connection and select Properties from menu.
Select TCP/IP	4. The Local Area Connection Properties window displays. In the General tab, click to highlight Internet Protocol (TCP/IP).
Properties	5. Click Properties to view TCP/IP properties.
IP Address : <gestimpc IP addr> Subnet Mask :	6. On the host computer at the scanner workstation, open a C Shell window. Using the " more /etc/hosts " command to record the IP addresses for gestimpc _____. <site_hostname>-tgate _____

255.255.252.0 Default Gateway <leave blank>	(used later) 7. On the STIM PC, if addresses do not match, enter the IP address assigned to “gestimpc” in the scanner hosts file and netmask (e.g. 255.255. 252 .0) reserved for the stimulus PC.
OK	8. Click OK in Local Area Connection Properties window.
Launch BrainWave PXE	9. On STIM PC desktop, double-click Launch BrainWave PXE . The monitor should display the idle screen. NOTE : If the “Idle” screen does not appear after one minute, please double-click on Launch BrainWave PXE again.

8.1.3 Load AW Software

If the customer has purchased BrainWave PA for AW, please load the appropriate software.

- For versions AW4.2 or AW4.2P, refer to Direction 5158951-100, BrainWavePA 1.5 for AW Installation/Service Manual.
- For versions AW4.3 or AW4.4, refer to Direction 5198939-100, BrainWavePA 1.6 for AW Installation/Service Manual.

8.2 Reload/Upgrade Software

Advanced Users	New Users
Exit PXE	1. On STIM PC, stop PXE and go to Windows Desktop, press [Shift] + [Esc].
Insert CD	2. Place CD for BrainWave Stimulus in the CD drive.
Launch .exe file	2. To navigate to CD content, on the computer desktop, double-click on the My Computer icon. 3. Click on CD drive DVD/CD-ROM Drive to view contents of the BrainWave Stimulus CD. 4. Locate the executable (.exe) file and double-click on it. This file will automatically run the reload/upgrade for BrainWave software. 5. Accept all defaults for the software install.
Remove CD	6. Once reload/upgrade is complete, remove BrainWave Stimulus CD .
Reboot computer	7. Restart STIM PC . Software will automatically launch BrainWave PXE. Software is functioning properly if the “Idle” screen is displayed.

8.3 Restore/Reimage Software

Restoring or reimaging the computer is usually done in event of disaster recovery. GE supplies a drive image DVD for disaster recovery needs. It has bootable Ghost software to help reload the operating system.

Advanced Users	New Users
Insert CD	1. Place bootable disk, HDx PC Platform Software with BrainWave, into DVD drive. 2. Window will display the following warning : “Warning ! This CD-ROM will destroy all data on the hard drive and load software to the original state.”
To accept restore, press [Enter] .	3. To accept restore, press [Enter]. 4. Once restore is complete, remove DVD.
To cancel restore, remove CD	5. To cancel restore, remove DVD.

9 Troubleshooting

See BrainWave troubleshooting document 2359814.